

Breaking the Neptunyl Barrier: Direct Access to Neptunium(IV) in Aqueous Solution via Polyoxometalate-Mediated Reduction and Stabilization

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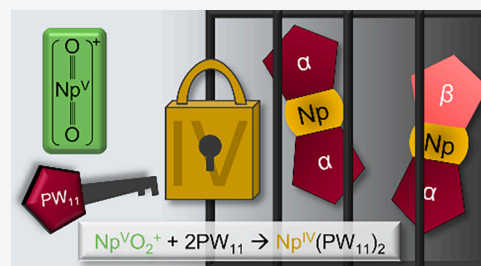
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ABSTRACT: Neptunium exhibits truly unique chemistry as its speciation is dominated by the neptunyl(V) ion (NpO_2^+). Here, we describe the spontaneous destabilization and reduction of neptunyl(V) via complexation to the Keggin-type polyoxometalate (POM) ligand $\text{PW}_{11}\text{O}_{39}^{7-}$. The POM-mediated reduction of NpO_2^+ does not require any reducing agent and occurs within minutes, at room temperature, and in aqueous solution. The resulting $[\text{Np}(\text{PW}_{11}\text{O}_{39})_2]^{10-}$ complex ($\text{Np}(\text{PW}_{11})_2$) remains soluble, water-stable, and air-stable for weeks and persists over an extended acidity range. Single-crystal structure, solid-state Raman and UV–visible absorbance characterization of $\text{Np}(\text{PW}_{11})_2$ revealed a mixed alpha/beta isomery of the Keggin ion, forming an unprecedented 50:50 mixture of $\text{Np}(\alpha\text{-PW}_{11})_2$ and $\text{Np}(\alpha\text{-PW}_{11})(\beta\text{-PW}_{11})$. Experiments with other tetravalent ions (i.e., Zr^{4+} , Hf^{4+} , Ce^{4+} , and Th^{4+}) indicate that the occurrence of the beta isomer is specific to Np^{4+} and independent of the cation's size. Solution-state characterization of the $\text{Np}\text{-PW}_{11}$ system via UV-visible-NIR absorbance, ^{31}P NMR, VT NMR, and relaxometry further elucidated the speciation. Moreover, comparative experiments with uranium revealed that the two types of actinyl ions (UO_2^{2+} vs NpO_2^+) undergo drastically different reactions in the presence of PW_{11} . UO_2^{2+} is not reduced but instead uses PW_{11} as a phosphate reservoir and precipitates as uranyl phosphate.



INTRODUCTION

Neptunium is truly a unique case in the periodic table as it is the only element whose chemistry is dominated by the pentavalent oxidation state and with an ion that exhibits two “yl” bonds, NpO_2^+ (Figure 1). While other oxidation states of Np can be obtained under specific conditions (from +II to +VII), most of the neptunium compounds that have been characterized contain neptunyl(V).¹ A recent comprehensive review from Gilson and Burns² on Np coordination chemistry accounted for a single structure for Np^{2+} , 16 for Np^{3+} , 91 for Np^{4+} , 149 for NpO_2^+ , and 91 NpO_2^{2+} , and for 18 NpO_4^- . Furthermore, most of the studies on neptunium have been focused on organic chelators (e.g., cyclopentadienyl-type ligands,^{3,4} crown ethers,⁵ arenes,⁶ carbenes,⁷ or porphyrins^{8,9}) or small inorganic anions (e.g., fluorides,¹⁰ hydroxides,¹¹ borates,¹² nitrides¹³). Most of the reported syntheses that involve oxidation states other than neptunyl(V) had to be performed under anhydrous, air-free, and/or low-temperature conditions and employed strongly reducing or oxidizing reagents.

In aqueous solution, Np most often forms the nearly linear pentavalent neptunyl ion, NpO_2^+ . With its two actinyl bonds and single positive charge, the NpO_2^+ has a low charge density and, as a result, it usually forms weaker complexes than the other f-elements.¹⁴ In contrast, Np^{4+} has a much higher charge

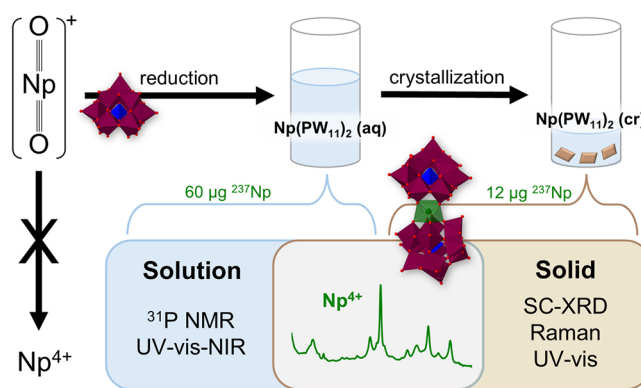


Figure 1. Overview of $\text{Np}\text{-PW}_{11}$ experiments, illustrating the POM-mediated reduction, solution speciation, and solid-state characterization. The Np reduction occurs overnight with 2 equivalents of PW_{11} in 0.1 M NaCH_3COOH adjusted to pH 5. Crystallization is induced with the addition of excess CsCl and occurs over the course of several days.

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density and forms strong complexes. Reduction to Np^{4+} requires breaking the strongly bonded neptunyl unit, i.e., a step which is often thermodynamically or kinetically unfavorable. It can be achieved through chemical reduction methods (e.g., excess iron sulfamate, hydroxylamine, and hydrogen peroxide), though these often proceed slowly, with low yields, or introduce components that interfere with subsequent chemistry (Table S1).^{15–19} Alternatively, neptunium can be reduced electrochemically,^{16,20–22} but such methods are not widely available and are unlikely to occur outside laboratory conditions.

In contrast to other chelator types, the use of polyoxometalates (POMs) in neptunium chemistry is largely underdeveloped. Out of the now ~600 Np compounds reported in the Cambridge Crystallography Database²³ and the NIST Inorganic Crystal Structure database,²⁴ only 9 are with POM ligands (Table S2). POMs exhibit distinct coordination geometry, redox behavior, and solubility properties relative to other types of chelators.²⁵ They also display an almost limitless range of compositions and structures (encompassing almost all blocks of the periodic table), making them useful to a wide variety of applications.^{25–27} While our earlier efforts to expand POM chemistry with transuranic elements focused on trivalent americium and curium^{28–32} and benefited from surrogate chemistry with trivalent lanthanides, neptunium had remained unexplored. Out of the few reports of Np-POM complexes,^{26,33–44} only four reveal the stabilization of Np^{4+} and they all required starting from prereduced precursors such as Np^{4+} oxalates, electrochemically reduced stocks, or Np^{4+} organometallic adducts (Tables S2 and S3).^{40–42,44} None the prior studies on Np-POM systems investigated the possibility of direct reduction from neptunyl.

In this work, we report the aqueous, one-step, and fast reduction of neptunyl(V) and long-term stabilization of Np^{4+} via a POM ligand. The tetradentate POM ligand $[\text{PW}_{11}\text{O}_{39}]^{7-}$ (PW_{11}) mediates the reduction of the NpO_2^{+} and forms a strong complex with Np^{4+} that is air-stable, water-stable, and remains intact over an extended period (>weeks) and over a broad acidity range (from pH ~5 down to 3 M HCl). The POM-mediated reduction is spontaneous and does not require any addition of reducing agent. The POM does not act as a reducing agent but instead as a thermodynamic redox potential shifter, which allows for the reduction of neptunyl(V), formation of a Np(IV)-POM complex, and concomitant oxidation of water. The Np- PW_{11} system was characterized in both aqueous solution and the solid state (Figure 1). The Np reduction was monitored by solution-state UV-vis-NIR spectrophotometry and the quantitative formation of $[\text{Np}(\text{PW}_{11})_2]^{10-}$ complex was confirmed. ^{31}P nuclear magnetic resonance (NMR) and variable temperature NMR measurements revealed additional levels of complexity for the Np-POM speciation in solution, with the presence of both α and β isomers of PW_{11} as well as inequivalent ^{31}P NMR signals. Relaxometry measurements were also performed, representing (to the best of our knowledge) only the second report of ^{31}P measurements on a Np compound.⁴⁵ We also report comparative NMR experiments with Zr^{4+} , Hf^{4+} and Th^{4+} . Furthermore, the $[\text{Np}(\text{PW}_{11})_2]^{10-}$ complex was crystallized at the microscale, using Cs^+ as a counterion, and characterized via single-crystal X-ray diffraction (SC-XRD). We also report analogous crystal structures with Zr^{4+} and Hf^{4+} which, combined with recently reported Ce^{4+} and Th^{4+} contrast with the uniqueness of neptunium chemistry.²⁷ Raman and

UV-vis measurements on the crystalline solids reveal further nuances in the Np^{4+} electronic structure due to rotational isomerism. Interestingly, control experiments with a smaller POM, $\text{W}_5\text{O}_{18}^{6-}$ (W_5), disclose that the neptunyl(V) reduction is specific to PW_{11} . Additionally, comparative experiments with uranyl(VI) (UO_2^{2+}) show no impact of PW_{11} on the redox chemistry of uranyl, opening possibilities for novel strategies with the *in situ* reduction of neptunyl and eventual recovery via POM-mediated selective precipitation.

EXPERIMENTAL SECTION

Materials. The following chemicals were used without further purification: sodium tungstate dihydrate (ACS grade, 99.0–101.1%, Alfa Aesar), sodium acetate (anhydrous, ACS grade, EM Science), cesium chloride (optical grade, >99.999%, Shelton Scientific), hydrochloric acid (6.00 N, RICCA), nitric acid (ACS reagent, 70%, Sigma-Aldrich), deuterium oxide (Aldrich, 99.9 atom % D), zirconium tetrachloride (Millipore Sigma), hafnium tetrachloride (Millipore Sigma), thorium nitrate hydrate (99.8%, Strem Chemicals) and uranyl nitrate hexahydrate (Spectrum Chemical, ACS reagent). Neptunium dioxide ($^{237}\text{NpO}_2$) from Lawrence Livermore National Laboratory's legacy material was dissolved and purified by anion exchange according to the method reported by Wang.⁴⁶ The resulting stock solution was 20 mM NpO_2^{+} in 0.3 M HNO_3 (Figure S1). All solutions were prepared using deionized water purified by reverse osmosis cartridge system ($\geq 18.2 \text{ M}\Omega\cdot\text{cm}$). All experiments were performed in a temperature-controlled room (18–22 °C).

Methods. CAUTION: ^{237}Np ($t_{1/2} = 2.144 \times 10^6$ years), depleted uranium (^{238}U $t_{1/2} = 4.463 \times 10^9$ years; ^{235}U $t_{1/2} = 7.04 \times 10^8$ years), and thorium (^{232}Th $t_{1/2} = 14.05 \times 10^{10}$ years) are alpha-emitting radionuclides. All samples were handled by trained workers in appropriate radiological facilities at Lawrence Livermore National Laboratory designed for the safe handling of long- and short-lived radioactive materials and associated waste.

Synthesis. POM solutions were made freshly for each experiment. PW_{11} was formed from the *in situ* conversion of a precursor, $\text{Na}_9\text{PW}_9\text{O}_{34}\cdot 16\text{H}_2\text{O}$ (PW_9) in 0.1 M $\text{Na}_3\text{CH}_3\text{COO}$ at pH 4.5–5.5.⁴⁷ PW_9 was synthesized in bulk according to the procedure previously reported by Contant et al.⁴⁷ W_5 forms *in situ* from the commercial salt $\text{Na}_2\text{WO}_4\cdot 2\text{H}_2\text{O}$ upon dissolution in the buffer matrix at pH ~5.⁴⁷ Np-POM samples were mixed by introducing an aliquot of the acidic neptunyl(V) stock into the NaCH_3COO buffer at pH ~5 with 2 equivalents of predissolved PW_{11} . Samples were prepared so that the Np concentration for the UV-vis spectrophotometry was 500 μM . For crystallization, the Np- PW_{11} solutions that had aged overnight (to ensure full Np reduction) were diluted to 100–200 μM Np and 800–3000 equivalents of Cs were added as a 6 M CsCl spike. Note, more concentrated Np reactions (e.g., 300 μM Np) rapidly precipitated upon Cs addition based on phase solubility. The precipitate could be redissolved with additional buffer for single crystal growth. Light brown crystals typically grew within 6 days. Zr-, Hf-, and Th- PW_{11} solutions and crystals were prepared similarly in the NaCH_3COO buffer at pH 4.5–5.5 for select comparative studies.

Single-Crystal X-ray Diffraction. Single crystals were mounted out of immersion oil onto MiTeGen MicroGrippers and contained within a MicroRT capillary (for radiological containment). Diffraction data were collected at room temperature on a custom Rigaku Synergy single-crystal X-ray diffractometer outfitted with a Kappa goniometer, MicroMax-007 HF microfocus rotating anode Mo X-ray source, and a Dectris Pilatus 3R (300 K–CdTe) detector. Based on pre-experiment screening, experiments were run as determined by the Laue group and collected for complete redundancy. Data underwent scaling, integration, and multiscan absorption correction in CrysAlisPro. Structure solutions were achieved with intrinsic phasing (SHELXT)⁴⁸ and least-squares refinement (SHELXL)⁴⁹ in ShelXle software package.⁵⁰ Isomeric disorder in the PW_{11} unit was split and modeled in parts. Several of the Cs^+ cations were also disordered and refined to

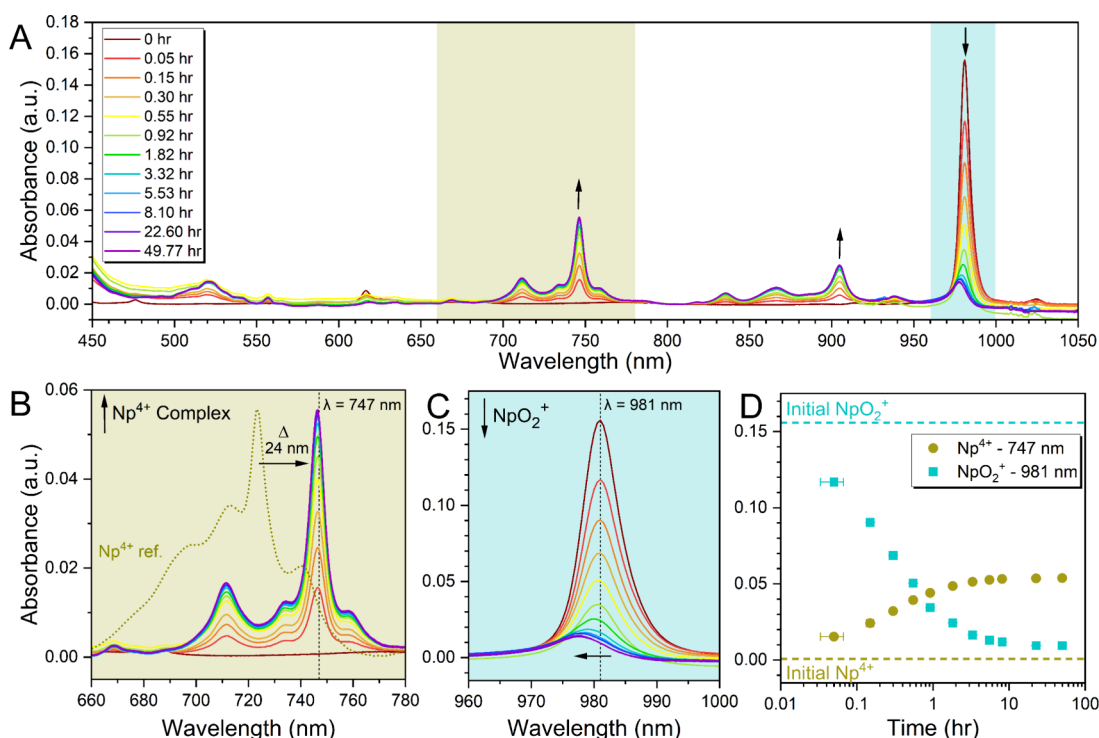


Figure 2. (A) Absorbance spectra of 0.5 mM Np with 2 equivalents of PW_{11} in 0.1 M sodium acetate buffer at pH ~ 5 , as a function of time. The tan and blue areas are further detailed in panels B and C. (B) Magnified inset of the Np^{4+} ingrowth and comparison to a Np^{4+} reference spectrum. The spectrum of the uncomplexed Np^{4+} has been normalized for clarity. (C) Magnified inset of the NpO_2^+ reduction. The arrows in panels A, B, and C indicate the direction of the main spectral changes as a function of time. (D) Reaction monitoring of the 747 nm (Np^{4+}) and 981 nm (NpO_2^+) absorbance values.

half occupancy. Lone oxygen atoms represent solvent water molecules, as the hydrogen atoms could not be located.

Solution UV-vis-NIR Spectrophotometry. Absorbance spectra were collected using an Agilent Cary 6000i UV-vis-NIR spectrophotometer. For Np samples, a minimum of 500 μL of sample was transferred to a sealed semi-micro quartz cuvette and scanned over 450–1050 nm with an average scan rate of 0.2 s/nm and 0.5 nm steps. Spectra were baseline corrected using the sodium acetate buffer (pH 4.5–5) and spectra were set to zero at 810 nm where there is minimal signal from sample. Gaussian peak fitting was performed using the Fit Peaks (Pro) function in OriginPro to track peak shifting.⁵¹ Uranium samples were analyzed in UV-transparent plastic cuvettes (BRAND). When more than a constant baseline subtraction was necessary (i.e., to decouple the scattering effects in the U- PW_{11} system—Figure S8), PreDICT software⁵² was used to model the baseline consistently across sample sets.

Solid-State UV-Vis Spectrophotometry. Crystals were isolated in immersion oil on glass slides for analysis. UV-vis data were collected using a CRAIC 508 PV microspectrophotometer outfitted on an optical microscope. The spectral window was 400–800 nm as accessible by the white light source of the instrument. We report an average of 200 scans with an integration time per scan optimized for each sampling area.

^{31}P NMR Spectroscopy. An equilibrated sample of 0.5 mM $\text{Np}(\text{PW}_{11})_2$ (containing 37 μg ^{237}Np) was mixed with 10% v/v D_2O , loaded into a 5 mm PTFE liner, and nested inside an NMR tube for radiological containment. All NMR spectra were collected on a 600 MHz Bruker AVIII spectrometer with a 5 mm HX probe. ^{31}P 1D spectra were collected using a 90-degree pulse (12.63 μs) with a 2.34 s recycle delay. Variable-temperature experiments were collected by letting the sample equilibrate at the set temperature for 10 min before data collection began. Between 512 and 1024 scans were collected at each temperature (283–338 K). After the heating and cooling of the sample was completed, a room temperature spectrum was collected to ensure that no irreversible chemical changes within the sample had

occurred. The temperature was externally calibrated using the residual proton signals of deuterated d_4 -methanol.⁵³ T_1 measurements at 298 K were completed using the inversion recovery sequence with interpulse delays ranging between 1 ms–10 s for short T_1 species and 100 ms–100 s for long T_1 species.²⁹ All ^{31}P spectra were referenced to the ^2H lock signal of a 90%/10% solution of H_2O and D_2O .

Raman Spectroscopy. Raman spectra were collected on crystalline solids with a Bruker Senterra II confocal Raman microscope. A 532 nm laser was operated at 12.5 mW power, the aperture was set to $50 \times 1000 \mu\text{m}$, 16 scans (per analyzed spot), 500 ms integration time (per scan), with automatic cosmic ray removal. Data were collected from 50–1435 cm^{-1} on multiple spots and representative average spectra are shown.

RESULTS AND DISCUSSION

Polyoxometalate-Mediated Reduction of Neptunyl(V) to Neptunium(IV) by PW_{11} . Like other actinides, neptunium exhibits sharp f - f transitions and characteristic solution colors that can be valuable to examine oxidation states and speciation.^{54,55} Therefore, UV-vis-NIR spectrophotometry was used to monitor the interactions between NpO_2^+ and the PW_{11} POM in aqueous solution. As prior studies had shown the possible complexation of neptunyl and plutonyl ions with lacunary POMs^{26,35,39,43} and others used reducing agents or electrochemistry to access Np^{4+} chemistry,^{40,41,56} we initially expected to observe the simple chelation of NpO_2^+ by PW_{11} , presumably forming hypothetical complexes such as $[\text{NpO}_2(\text{PW}_{11}\text{O}_{39})_2]^{13-}$ or $[\text{NpO}_2(\text{PW}_{11}\text{O}_{39})]^{6-}$. Unexpectedly, mixing neptunyl(V) and PW_{11} in aqueous solution, without adding any reducing agent and under ambient conditions, led to the rapid reduction of NpO_2^+ and formation of Np^{4+} species.

Upon introduction of neptunyl(V) into an aqueous solution of PW_{11} , with 2 stoichiometric equivalents, $\sim 25\%$ reduction of the characteristic NpO_2^+ peak at 981 nm is evident within 3 minutes, $\sim 78\%$ reduction after 1 hour, and the Np continues to reduce until a redox steady state is achieved after equilibration overnight (Figure 2). Concurrently, peaks at 712, 747, 835, 867, and 905 nm grow in. These spectral bands are indicative of Np^{4+} .⁵⁴ The band at 747 nm exhibits a bathochromic shift of 24 nm relative to the free Np^{4+} -aquo complex (723 nm, Figure 2B), a consequence of the strong ligand imposed by the PW_{11} anions. This shift is consistent with what has been observed for the Wells-Dawson type POM complex $[\text{Np}(\text{P}_2\text{W}_{17}\text{O}_{61})_2]^{10-}$ obtained by electrochemical preparation of Np^{4+} , reported by Chiang et al.⁵⁷ Note that our results were obtained at room temperature, in sodium acetate buffer at pH ~ 5 , and without any particular precaution regarding natural oxygen dissolved in the solution. It is also worth noting that the samples contained just 0.5 mM of the complex (0.5 mM Np + 1 mM POM), and faster reaction times are to be expected at higher concentrations or with excess POM. The reproducibility of the results was also verified (Figure S2).

The shift in the Np^{4+} peak is evidence of complexation with PW_{11} and has been seen in other *f*-element POM complexes,^{28,41,57} while the lack thereof in the NpO_2^+ peak, indicates the complexation only occurs after reduction. Attempts to model the reaction rate did not yield any obvious kinetic fits with first- or second-order reactions, which supports a multistep mechanism. The transdioxo NpO_2^+ ion only accommodates equatorial ligands^{2,58} and is not compatible with the square plane of the four O atoms in the lacunary heteropolytungstate Keggin ion (Figure S3). The lingering band at 977 nm (9.2% of initial) in the steady-state spectrum could, at first sight, indicate incomplete reduction due to the decreasing concentrations. However, we also note that Np^{4+} also has a characteristic peak at 960 nm so this peak could be shifted to 977 nm upon complexation. A Np^{4+} band at 980 nm has been previously reported for high nitrate (>2.5 M) media.^{55,59} By comparing the 981 nm peak decrease (Figure 2D) and the 981 nm peak shift (Figure S4), we observe different trend concavities, which further supports the lingering signal at 977 is not simply incomplete reduction but more likely an Np^{4+} band affected by the POM's ligand field.

A closer look at the lower energy region (>1050 nm) reveals a minor peak at 1221 nm (Figure S5). Neptunyl(V) is known to disproportionate into Np^{4+} and NpO_2^{2+} in the absence of ligands that favor one its oxidation states. In the presence of PW_{11} , the equilibrated samples (i.e., aged overnight) exhibit a faint peak at 1218 nm with a different shape than what is traditionally observed for NpO_2^{2+} . The peak is also blue-shifted related to the reference peak of NpO_2^{2+} (1218 vs 1221 nm) which makes it unlikely that this peak would be due to a POM complex with NpO_2^{2+} . Therefore, we interpret the minor peak at 1218 nm as a peak of Np^{4+} that is normally observed at 1156 nm when uncomplexed (Figure S5) but is red-shifted to 1218 nm upon complexation and stabilization by PW_{11} . Overall, the spectral features in this region are challenging to resolve due to the noise level. In following sections, we detail a thermodynamic driving factor for the formation of the Np^{4+} -POM complex, and solution-state NMR data further support that only Np^{4+} persists in the equilibrated POM sample.

Regardless, over 90% of the NpO_2^+ has been reduced to Np^{4+} by simple addition of the POM and thus constitutes the

bulk of the solution speciation. Such complexation competes with the typical hydrolysis behavior of tetravalent actinides at pH > 1 .⁶⁰ Given the 2:1 ratio POM:Np used in these experiments and prior compounds reported with Keggin-type POMs, including with Th^{4+} and U^{4+} ,^{27,61} we posited that the resulting complex would be $[\text{Np}(\text{PW}_{11}\text{O}_{39})_2]^{10-}$ (abbreviated $\text{Np}(\text{PW}_{11})_2$). Figure S6 shows the UV-vis-NIR spectrum of $\text{Np}(\text{PW}_{11})_2$ compared to that of Np^{4+} colloids. While the spectral features have some similarity around 740 nm due to the Np^{4+} peak shifting by complexation, the sharp character contrasts with the typical broad bands of poorly ordered Np^{4+} hydrolysis products. Thus, the $\text{Np}(\text{PW}_{11})_2$ complex is more thermodynamically favorable than hydrolysis products in these conditions, resulting in POM-induced solubilization of Np^{4+} , even at pH 5 and at 0.5 mM concentration (i.e., $[\text{Np}] \approx 120$ mg/L). Remeasuring solutions that aged for a month did not reveal any reoxidation back to NpO_2^+ (Figure 3A), even upon

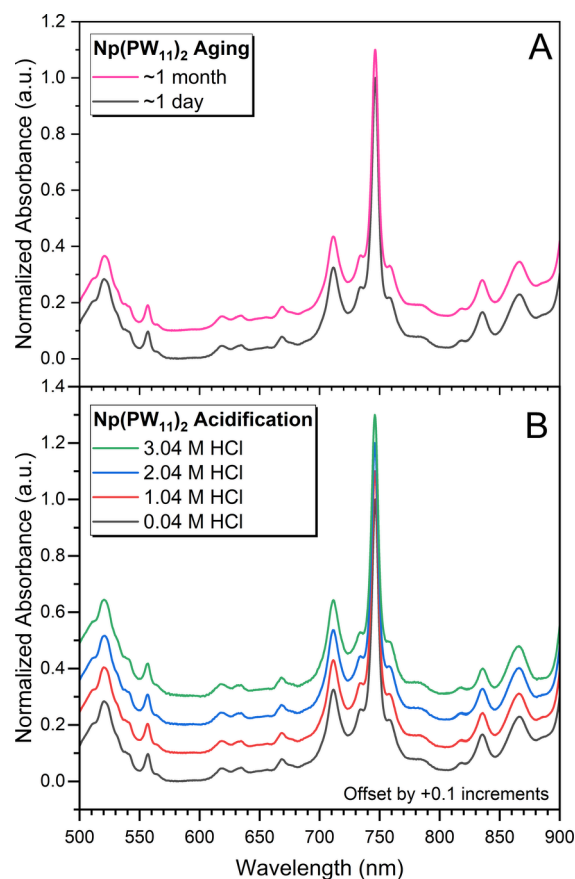


Figure 3. UV-vis spectra of the $\text{Np}(\text{PW}_{11})_2$ solution upon (A) aging and (B) acidification with HCl. No peak shift for Np was observed. $[\text{Np}]_{\text{total}} = 0.5$ mM. $[\text{POM}]_{\text{total}} = 1$ mM. The HCl concentration corresponding to each curve is indicated in the legend.

storage at room temperature and in a fumehood (sealed sample but exposed to air). Furthermore, titrations of the Np - PW_{11} reaction solution with HCl reveal that the complex persists upon acidification to 3 M HCl (Figure 3B), as indicated by the lack of peak shifting and stable spectral features consistent with Np^{4+} .

Rationalizing the POM-Mediated Reduction Neptunyl(V) to Neptunium(IV). Observing the spontaneous reduction of NpO_2^+ to Np^{4+} in aqueous solution, upon simple addition of PW_{11} appeared, at first, surprising. Indeed, NpO_2^+

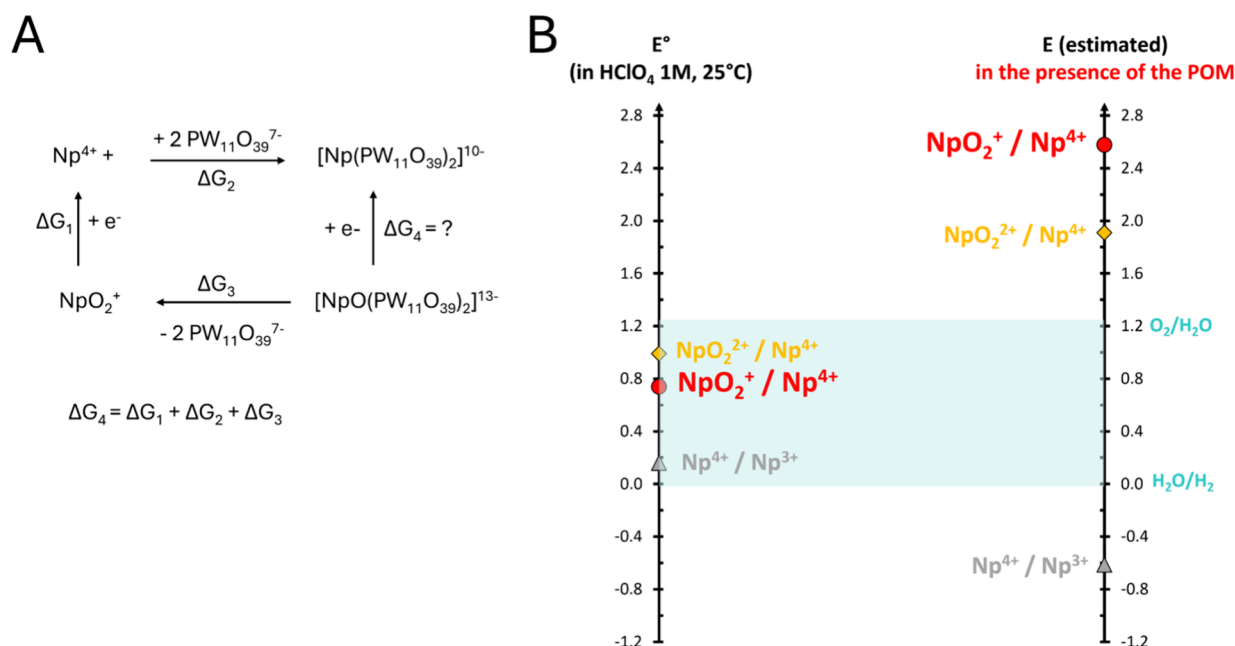


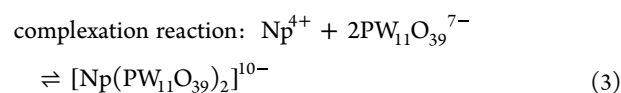
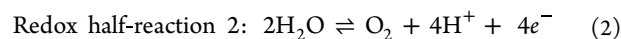
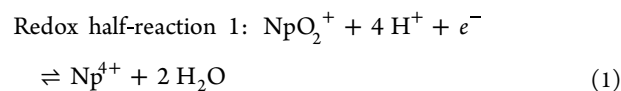
Figure 4. (A) Thermodynamic cycle involving the reduction of neptunyl(V) to neptunium(IV) with and without complexation to PW_{11} . The Nernst equation and the complexation reactions allow for the estimations of the unknown ΔG and hence the associated redox potential. (B) Tabulated redox potentials of the neptunium couples under standard conditions and estimates for the same Np couples upon complexation to PW_{11} . Details of estimations given in Table S4. The blue area delineates the redox window in water.

is reputedly a stable ion, and it traditionally requires a reducing agent or water-free or air-free conditions to be reduced. The POM ($\text{PW}_{11}\text{O}_{39}^{7-}$) is also a stable species and is fully oxidized, composed of WO_4^{2-} octahedra and a PO_4^{3-} ion. Therefore, even though the reaction only happens once the POM is introduced in the system, the POM cannot act as a reducing agent for NpO_2^+ . Instead, the observed NpO_2^+ to Np^{4+} reduction is likely the result of a thermodynamic redox potential shift induced by the ligand complexation (Figure 4). Significant redox shifts are not uncommon for other metals, with, for example, shifts larger than 1 V for the $\text{Ce}^{4+}/\text{Ce}^{3+}$ couple.^{62–64} Some redox shifts for the neptunium redox couples in acidic media and in the presence of another POM, $\text{P}_2\text{W}_{17}\text{O}_{61}^{10-}$, have also been reported in solution-state studies by Shilov et al.³⁷ and by Antonio et al.⁵⁷ If the ligand in question has a strong thermodynamic selectivity for one cation over the other, then the Nernst equation dictates that the apparent redox potential will shift toward formation of the strong complex. Such a mechanism can trigger the redox reaction without the need of a reducing or oxidizing agent as the solvent itself can act as reduced or oxidized byproduct. In the case of aqueous systems, if the redox potential is shifted higher than that of the $\text{O}_2/\text{H}_2\text{O}$ couple ($E^\circ = +1.23$ V), then water can take part in the reaction.

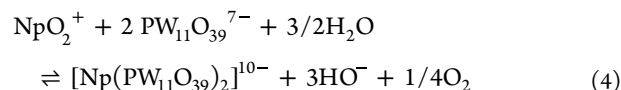
In the case of POMs, Yusov et al. determined the complexation constant of $\text{P}_2\text{W}_{17}\text{O}_{61}^{10-}$ with Np^{4+} , measuring a $\log \beta_{12}$ of 35.0.³⁸ The same POM binds Am^{3+} with a $\log \beta_{12}$ of 21.9³⁸ and binds uranyl (UO_2^{2+}) in a much weaker way with a $\log \beta_1$ of just 3.9.⁶⁵ Knowing that NpO_2^+ ions have a lower charge density than UO_2^{2+} , an even lower binding constant can be expected for neptunyl(V) with the POM. This is consistent with our UV–vis absorbance results which show no evidence of POM complex with neptunyl(V). While the $\text{P}_2\text{W}_{17}\text{O}_{61}^{10-}$ POM is not the perfect surrogate for $\text{PW}_{11}\text{O}_{39}^{7-}$, both POMs are lacunary phosphotungstates, act as tetradentate ligands, and have the same coordination moieties. Thus, the thermody-

amic data on $\text{P}_2\text{W}_{17}\text{O}_{61}^{10-}$ can be used to provide a coarse estimate of the stability of the $\text{PW}_{11}\text{O}_{39}^{7-}$ complexes (Table S4). Figure 4B and Table S4 provide the estimated redox potentials for the different neptunium couples in the presence of the POM. The $\text{NpO}_2^+/\text{Np}^{4+}$ potential is increased by +1.8 V bringing the NpO_2^+ species outside the stability window of water. As such, if the initial system contains NpO_2^+ and the POM, water becomes a species that can act as a reducing agent, leading to the Np^{4+} complex with $\text{PW}_{11}\text{O}_{39}^{7-}$. While the redox potentials shown in Figure 4B must be taken as coarse estimates, the redox shift the $\text{NpO}_2^+/\text{Np}^{4+}$ couple appears large enough that we can anticipate with confidence that the actual value would be higher than that of the $\text{O}_2/\text{H}_2\text{O}$ couple. These calculations also rationalize why the neptunyl reduction in the presence of PW_{11} occurs spontaneously in aqueous solution, why the reduction stops at Np^{4+} , and why the Np^{4+} complex is not reoxidized.

The following scheme summarizes the reactions involved in the spontaneous reduction of NpO_2^+ and stabilization of Np^{4+} in aqueous solution via complexation to the POM.



Sum reaction:



Crystal Structures of $\text{Np}(\text{PW}_{11})_2$ and Analogous Zr, Hf, Ce, and Th Complexes. Crystals of the $\text{Np}(\text{PW}_{11})_2$ complex were grown using excess CsCl as a crystallizing agent on samples that had aged at least overnight for equilibrated Np reduction. Most of the crystals were irregularly shaped and unsuitable for single-crystal X-ray diffraction, though their light brown color was used to distinguish them from other non- Np bearing colorless simple salts (Figure S7). However, rhombohedral single crystal fragments were isolated from crystalline aggregates for analysis. Refinement of the XRD results revealed the compound as a sandwich-type complex, with two tetradentate POM ligands coordinating one central Np^{4+} ion (Figure 5). This new Np compound, fully formulated

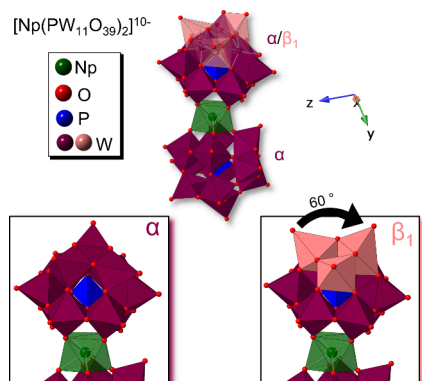


Figure 5. Crystal structure of the $\text{Np}(\text{PW}_{11})_2$ complex. Neptunium, phosphate, and tungstate coordination polyhedra are represented in green, blue, maroon/salmon, respectively. Oxygen is shown as red spheres of arbitrary radius. Charge balancing Cs^+ and Na^+ cations and solvent water molecules are omitted for clarity. Full formula of the compound: $\text{Na}_{4.5}\text{Cs}_{5.5}[\text{Np}(\text{PW}_{11}\text{O}_{39})_2] \cdot 11\text{H}_2\text{O}$. The disordered model is split into the alpha and beta isomer parts. See Figures S8 and S10 for additional view of the structure. See Tables S5 and S6 for additional crystallographic data on $\text{Np}(\text{PW}_{11})_2$.

$\text{Na}_{4.5}\text{Cs}_{5.5}[\text{Np}(\text{PW}_{11}\text{O}_{39})_2] \cdot 11\text{H}_2\text{O}$, crystallizes in the low symmetry, triclinic $P-1$ space group (Table S5). The asymmetric unit contains one crystallographically unique $\text{Np}(\text{PW}_{11})_2$ complex completed with solvent water molecules and disordered charge balancing Cs^+ and Na^+ ions (Figure S7).

While Cs^+ counterions clearly take part in the crystallization process, no additional complexation-induced peak shifting was observed in the solution-state UV–vis spectrum upon addition of CsCl (Figure S9). Each unit cell contains two $\text{Np}(\text{PW}_{11})_2$ complexes and the associated interstitial species. The Np centers are coordinated to 8 oxygen atoms (two tetradentate PW_{11} ligands on each side of Np) and have square antiprismatic geometry. The Np – O bond distances range 2.30–2.38 Å and are consistent with previous Np^{4+} –POM compounds obtained with reducing agents (Table S2).^{40,41,44,56} The neptunium oxidation state is further confirmed as Np^{4+} with a bond valence value of 4.00 (Table S6). Positional disorder in one of the PW_{11} units reveals the presence of two different rotational isomers of the Keggin POM. The plenary Keggin POM, PW_{12} , is known to possess α , β , γ , δ , ϵ rotational isomers, characterized by the successive rotation of its four trimeric $(\text{WO}_6)_3$ subunits.⁶⁶ These isomers persist for the lacunary PW_{11} Keggin ion, but the more symmetrical α – PW_{11} is most common and typically the most thermodynamically stable.

We here observe the 60° rotation of one trimeric subunit to yield a β isomer (Figure 5). By allowing the occupancy of the disordered W trimers to refine as free variables, we observe near even contributions from the α and β isomers with 51% occupancy for the W20–W21–W22 trimer and 49% occupancy for the W23–W24–W25 trimer, respectively. A closer look at residual electron density on the other PW_{11} unit indicates a small fraction of the β isomer (<10% occupancy) may be present at the W8–W9–W11 trimer, but this is unresolvable for the available data set. It can be reasonably concluded that the $\text{Np}(\text{PW}_{11})_2$ compound isolated here is overall a 75:25 mixture of the α and β isomers (one fully α complex, plus one 50:50 α/β complex). This may result from a decrease in the energetic barrier between the two isomers upon complexation to Np^{4+} . It is worth noting that isolating the β isomer of a phosphotungstate POM is rare. Weinstock and co-workers⁶⁷ have previously shown via solution-state NMR that, within the series of Keggin ions $[\text{XW}_{12}\text{O}_{40}]^{n-}$ (with $\text{X} = \text{P}^{5+}$, Si^{4+} , or Al^{3+}), $\text{PW}_{12}\text{O}_{40}^{3-}$ offers the least favorable structure for the β isomer. These results were later confirmed via DFT calculations by Poblet et al.,⁶⁸ showing that the β isomer of $\text{PW}_{12}\text{O}_{40}^{3-}$ is thermodynamically unstable, with an energy difference between α and β isomer of +6.5 kcal/mol.

Here we also performed the synthesis with Zr^{4+} and Hf^{4+} and report the isostructural crystal structures for $\text{Zr}(\text{PW}_{11})_2$ and $\text{Hf}(\text{PW}_{11})_2$ (Figure S8 and Table S5). Both $\text{Zr}(\text{PW}_{11})_2$ and $\text{Hf}(\text{PW}_{11})_2$ crystallize as 100% α isomer. When combined with the corresponding structures we previously reported^{27,28} with Ce^{4+} and Th^{4+} , this yields a consistent series to contrast the neptunium case. Other $\text{M}(\text{PW}_{11})_2$ compounds were also previously reported for a wide variety of central cations outside of the f -block.^{69–75} In all cases except Np , only the α isomer was obtained. For further contextualization, the ionic radius of Np^{4+} falls within that of Ce^{4+} and Th^{4+} .⁷⁶ Interestingly, within the series of tetravalent $\text{M}(\text{PW}_{11})_2$ structures, where $\text{M} = \text{Hf}^{4+}$, Zr^{4+} , Ce^{4+} , Th^{4+} , and Np^{4+} , there exists a near perfect correlation between the M – O distances and the ionic radius of the central cation (Figure 6A). This further supports the tetravalent Np oxidation state for the $\text{Np}(\text{PW}_{11})_2$ compound. The results also indicate that the existence of the beta isomer within $\text{Np}(\text{PW}_{11})_2$ is not due to a size constraint.^{27,76} Notably, the Np complex presents a strong bent, with 162.5° angle PW_{11} – Np – PW_{11} axis (Figure 6B). For comparison, both the Zr and Hf structures have a bent angle of just 167.2°. Such axial bending has been observed in other $\text{M}(\text{PW}_{11})$ compounds and is a distinct feature for P compared to other heteroatoms in the $\text{XW}_{11}\text{O}_{39}^{n-}$ series of POMs (where $\text{X} = \text{B}^{3+}$, P^{5+} , Si^{4+} , Ge^{4+} , Ga^{3+}).²⁷ Strong bending is consistent with a larger torsion angle (36.2°, twisted 8.8° from the idealized 45° position) and shorter heteroatom distances ($d(\text{P}$ – $\text{P})$ 8.766 Å).^{27,32}

To the best of our knowledge, the only case in the periodic table that has been isolated as 100% β isomer with PW_{11} is when it is complexed to curium (Cm^{3+} in $\text{Cs}_{11}\text{Cm}(\beta\text{-PW}_{11}\text{O}_{39})_2 \cdot 21.5\text{H}_2\text{O}$).²⁸ The $\text{Np}(\text{PW}_{11})_2$ compound therefore remains a unique case that seems to adapt a behavior that is intermediate between that of a lighter tetravalent metal and that of a heavy actinide. Moreover, monolacunary β isomers are further subcategorized as β_1 , β_2 , and β_3 based on the location of the rotated trimer relative to the lacunary position, where the central metal coordinates.⁷⁷ The $\text{Np}(\text{PW}_{11})_2$ structure is β_1 -type; the rotated trimer is separated from the central metal by a tungstate unit. The previously reported β

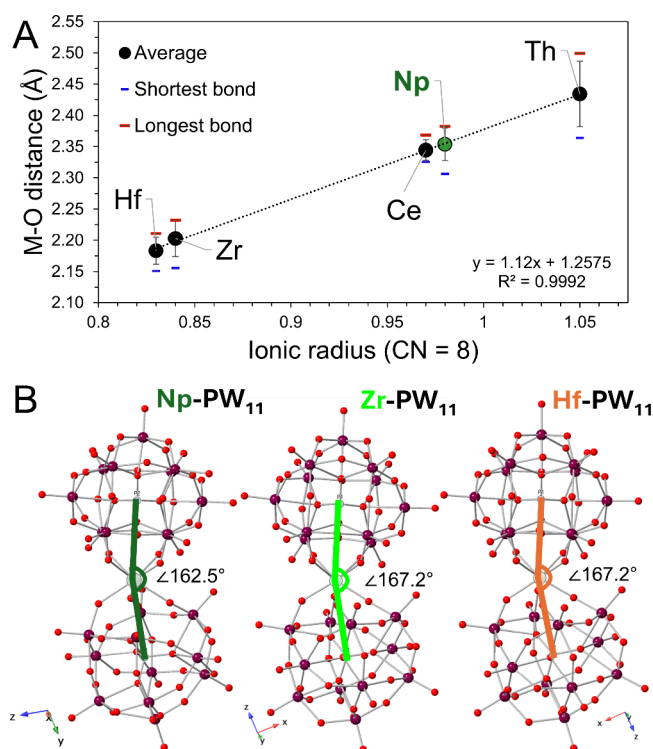
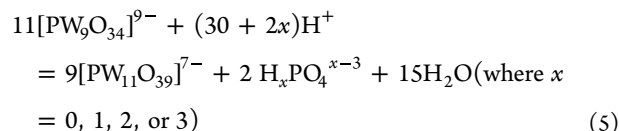


Figure 6. (A) Average M–O distances as a function of the tetravalent metal center ionic radii for the $M(PW_{11})_2$ crystal structure series, where $M = Hf^{4+}$, Zr^{4+} , Ce^{4+} , Np^{4+} , and Th^{4+} . (B) Ball-and-stick model of $M(PW_{11})_2$, where $M = Np^{4+}$, Zr^{4+} , Hf^{4+} , to illustrate the axial bending by $\angle P1-M-P2$.

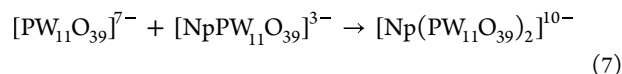
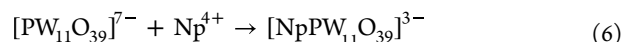
isomer of $Cm(PW_{11})_2$ exhibits the β_2 -type, whereby the rotated trimer is coordinated to the central metal.²⁸ Mixed α/β_2 isomerism has been reported previously in germanopolytungstates compounds $K_{13}M(GeW_{11}O_{39})_2 \cdot nH_2O$ (where $M = Ho^{3+}$, Er^{3+} , and Tm^{3+}) but not for PW_{11} .⁷⁸ It is unclear as this point whether the emergence of the stabilization of the two isomers of PW_{11} upon binding to Np^{4+} is due to the reaction pathway we chose in this study (with a reduction from NpO_2^+) or a specificity of tetravalent actinides. Experiments are underway to investigate plutonyl(V) and plutonium(IV) chemistry with the POM.

^{31}P NMR Spectroscopy and Relaxometry of the Np^{4+} - PW_{11} System. To further navigate the solution speciation, we conducted ^{31}P NMR on the steady-state $Np(PW_{11})_2$ solutions. ^{31}P NMR is a powerful tool to distinguish speciation for PW_x -containing samples^{79,80} and we leveraged the high sensitivity to comply with radiological constraints inherent to transuranic elements, analyzing a sample containing merely 37 μg ^{237}Np (~ 52 nCi, including the short-lived ^{233}Pa decay product). Room-temperature collection reveals six ^{31}P signals at 3.40, -7.46 , -17.94 , -18.93 , -20.90 , and -21.93 ppm (Figure 7). The most downfielded signal at 3.40 ppm (peak A) is from unbound phosphate in the system due to the stoichiometric excess in the PW_9 to PW_{11} conversion (eq 5).



The unbound PW_{11} POM is at -7.46 ppm (peak B) as previously reported.²⁹ There are four peaks shifted 10–15 ppm upfield from the free POM and ascribed to the Np complexes.

Previously, we identified the presence of 1:1 and 1:2 complexes of PW_{11} with cations of lower charge density (i.e., trivalent lanthanides as well as Am^{3+} and Cm^{3+}).²⁹ However, in the $Np-PW_{11}$ system, there is no evidence for multiple complexes in the absorbance spectra (Figure 2) so we anticipated that one of two possible species (either the 1:1 or 1:2 complexes, eqs 6 and 7) would be predominant.



The charge density of the tetravalent cation in solution, relative to trivalent cations, should drive the formation of the 1:2 complex, as tetravalent cations tend to form stronger complexes with hard bases, like the POMs.⁸¹ Furthermore, stopping the $Np-PW_{11}$ complexation scheme at the 1:1 species would leave Np^{4+} exposed to multiple solvent molecules (as its first coordination sphere typically requires 8 neighbors) which would certainly lead to hydrolysis and/or precipitation. Under these conditions, and considering the crystal structure

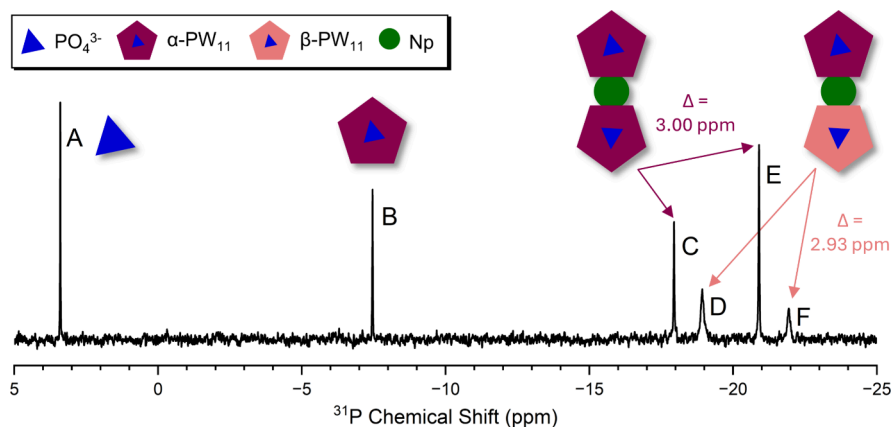


Figure 7. ^{31}P NMR spectrum of the steady-state $Np(PW_{11})_2$ solution, annotated with the respective species. Letter designations indicate different species further discussed in the text. See Table S7 for peak positions, peak areas, and corresponding T_1 values. See Figure S12 and Table S8 for comparative results of PW_{11} samples with other tetravalent metals, $Zr(PW_{11})_2$, $Hf(PW_{11})_2$, and $Th(PW_{11})_2$.

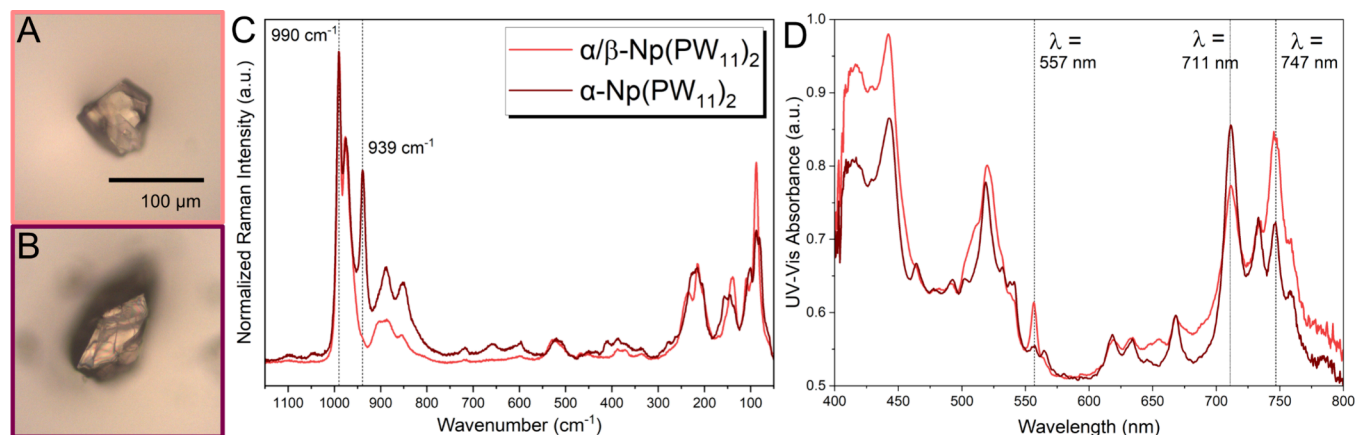


Figure 8. Microscopy images of crystal fragments of (A) α/β -Np(PW₁₁)₂ and (B) α -Np(PW₁₁)₂. (C) Solid-state Raman (C) and solid-state UV-vis (D) spectra for the corresponding Np(PW₁₁)₂ isomeric structures. The legend is the same for (C) and (D).

obtained, the 1:2 species (eq 7) is therefore the most likely species in solution for the Np⁴⁺-PW₁₁ system.

Reinforcing this hypothesis, previous work from Kato et al.⁷¹ with ethylamine salts of the [Zr(PW₁₁)₂]¹⁰⁻ and [Hf(PW₁₁)₂]¹⁰⁻ complexes dissolved in D₂O revealed a simple speciation distribution with only the 1:2 complex observed. However, the ³¹P spectra of the Zr-PW₁₁ and Hf-PW₁₁ samples displayed NMR doublets, separated by just ~0.1 ppm, corresponding to two inequivalent P signals.⁷¹ To benchmark the behavior of Np⁴⁺, we prepared similar solutions of M(PW₁₁)₂ (where M = Zr⁴⁺, Hf⁴⁺, and Th⁴⁺) in a matrix consistent with our other experiments (0.1 M sodium acetate, pH ~5). For all three diamagnetic ions (Zr⁴⁺, Hf⁴⁺, and Th⁴⁺), we observed a doublet peak (Figure S12), separated by 0.11, 0.13, and 0.15 ppm, respectively. This is consistent with and extended the observation made by Kato et al.⁷¹ We posit that the tetravalent metals allow for the observation of the two inequivalent P in the [M(PW₁₁)₂]¹⁰⁻ complexes because of their high thermodynamic stability,⁸¹ which prevents exchange with the solvent or free POM. The slight structural differences between the two PW₁₁ ligands around the central cations render them inequivalent at the NMR time scale. In the case of Np(PW₁₁)₂, the spectrum (Figure 7) is further complicated by the presence of the α and β isomers of PW₁₁ (Figure 5), which should result in a pair of doublet peaks. Furthermore, we expected the paramagnetic nature of Np⁴⁺ to enhance the peak separation between the two inequivalent ³¹P NMR signals (for each doublet). Given the similar sharpness, we assign peaks C/E as a doublet corresponding to the α isomer. The two broader peaks, D/F, are likewise a doublet of the mixed α/β isomer. The relative broadness can either be due to additional configurations and anisotropy in the β versus α isomers or due to differences in the spin density around the phosphate sites of the two isomers. The peak separation for both doublets (C/E and D/F) is ~3 ppm. This is much higher than the 0.11–0.15 ppm that we observed for the Zr, Hf, and Th systems (Table S8), but it is correlated by increasing Z and the paramagnetism of Np⁴⁺ (configuration 5f³). Integrating the signals reveals the pure α and the mixed α/β species are near evenly distributed in solution with the former comprising 46% and the latter comprising 54% of the Np species (Table S7). This relative α and α/β isomeric distribution in solution matches with the average 75% α and 25% β occupancies observed in the crystal structure. While Np⁴⁺ can stabilize the

β -PW₁₁ isomer, the α isomer still appears more stable. This would be congruent with the results from Kortz et al. on the β isomers of the [Ln(SiW₁₁O₃₉)₂]¹³⁻ and [Ln(GeW₁₁O₃₉)₂]¹³⁻ complexes.^{77,78}

Nuclear spin–lattice relaxation time (*T*₁) data were collected for each ³¹P signal and are reported in Table S7. The Np species have short relaxation times (*T*₁ = 95–252 ms), owing to the paramagnetic effects from the Np proximity to the P nucleus (distance of 4.44 Å). These *T*₁ values are similar to those reported for other paramagnetic complexes in solution, [Nd(PW₁₁O₃₉)₂]¹¹⁻ and [Eu(PW₁₁O₃₉)₂]^{11-,29}. The short relaxation times further support full reduction of NpO₂²⁺ to Np⁴⁺ in solution instead of lingering disproportionation. Diamagnetic neptunyl(VI), NpO₂²⁺, would have significantly longer ³¹P *T*₁ values, closer to that of the uncoordinated POM (18.1 s), and its chemical shifts would also be close to the unchelated POM. We undertook a variable-temperature study from 283–338 K on the Np(PW₁₁)₂ system (Figure S13). The free phosphate and PW₁₁ peaks (A and B), reveal a monotonic downfield shift with increasing temperature. Conversely, the Np-PW₁₁ species have nonlinear behavior with temperature (Figure S14), complicated by the dynamic exchange between different species. Generally, the abundance of the β isomer is reduced with elevated temperatures, likely owing to the higher stability of the α isomer.

Solid-State Optical and Vibrational Spectroscopy of the Np-PW₁₁ System. To further confirm the formation of the isomers, crystalline fragments of Np(PW₁₁)₂ were analyzed via solid-state Raman (Figure 8C) and solid-state UV-vis (Figure 8D) spectroscopies. As shown in optical microscope images, the two isomers were morphologically indistinguishable (Figure 8A,B) yet exhibit distinct spectral properties. In the Raman spectra, the most dominant feature arises from the stretching of the terminal W=O bonds (920–1010 cm⁻¹). Band shifting in this region can be used to distinguish the lacunary complexation of the Keggin ligand (990 cm⁻¹) from that of its Na-PW₉ precursor (971 cm⁻¹) (Figure S16). Furthermore, the presence of the peak at 939 cm⁻¹ is the signature of the pure α isomer. We previously identified this feature in the Raman spectrum of Cs_xM(PW₁₁)₂·nH₂O containing the α isomer (M = Am³⁺, Ln³⁺, Ce⁴⁺, Th⁴⁺).^{28,31} The 939 cm⁻¹ signal is also visible as a peak shoulder in aggregated crystalline solids of Np(PW₁₁)₂ that are likely a mixture of isomers (Figure S16). Multiple weaker bands in the

830–920 cm^{-1} region correspond to the $\text{W}_2\text{-}\mu_2\text{O}$ and Np-O-W stretches. The $\text{W}_3\text{-}\mu_3\text{O}$ stretches are in 490–550 cm^{-1} region. Peaks below 300 cm^{-1} are more difficult to assign, although Thouvenot et al.⁸² have previously noted it is the best spectral region to distinguish isomerism in the full PW_{12} Keggin ion; this does not seem to be the case for the coordinated lacunary Keggin structures.

Further evidence for the rotational isomers manifests in the solid-state UV–vis spectra of the $\text{Np}(\text{PW}_{11})_2$ crystalline materials in which an intensity inversion of the peaks in the peak envelope of Np^{4+} in the 700–750 nm region occurs (Figure 8D). A similar inversion has been observed when moving from the Np^{4+} -aquo to the Np^{4+} -nitrate complexes with increasing nitric acid concentration, though the exact species and coordination are still unknown.⁵⁵ The pure α crystal has the 711 nm peak as most dominant, while the mixed $\alpha_{0.75}/\beta_{0.25}$ crystal has the 747 nm peak as most dominant in that region. The mixed isomer also has a stronger feature at 557 nm. Comparison to the solution-state data (Figure S17), which has near equivalent α/β contributions, reveals that the intensity of the 747 nm peak (as normalized to the 711 nm peak) becomes stronger with greater β content.

Contrasting Neptunyl and Uranyl Reactions with POMs. Finally, to investigate the exclusivity of the Np-PW_{11} system, we undertook parallel experiments with another tetradentate tungstate POM, $\text{W}_5\text{O}_{18}^{6-}$ (W_5), and uranyl(VI) (UO_2^{2+}), exchanging both the POMs (W_5 and PW_{11}) and the actinides (NpO_2^+ and UO_2^{2+}), respectively. Like the experiments described above, UV-vis-NIR spectra were monitored for 0.5 mM neptunyl(V) with 2 equivalents of W_5 in 0.1 M sodium acetate buffer at pH 5 as a function of time (Figure S18). Contrasting with the behavior of the Np-PW_{11} system, only $\sim 3.2\%$ of the NpO_2^+ reduced over the course of 13 days. Partial reduction is evident by a very small decrease in intensity of the ~ 981 nm band (Figure S18C) and a corresponding increase for the 747 nm band (Figure S18B). We also tried exposing the sample to UV light (254 nm) to investigate whether photochemical mechanisms could accelerate the reaction progress (Figure S19), as was previously observed for some actinyl systems.^{83,84} Interestingly, the UV exposure reversed the minor NpO_2^+ -to- Np^{4+} reduction progress. The rather slow kinetics and almost negligible neptunyl(V) reduction in the Np-W_5 system as compared to Np-PW_{11} support that the central phosphate unit as playing a role in the POM-mediated neptunyl reduction. However, previous experiments by Antonio and Chiang⁸⁵ with electrochemically reduced Np^{4+} and a different phosphorus-containing POM, $[\text{P}_5\text{W}_{30}\text{O}_{110}]^{15-}$, reoxidized to NpO_2^+ and no Np-POM complex was formed. Likely, a steric component, the thermodynamic affinity of POM for the central metal, as well as the coordination mode of the POM are involved in the POM-mediated redox chemistry.

Similar experiments were conducted with uranyl(VI) and PW_{11} to test whether PW_{11} could also reduce and complex with uranium (Figure S20). Using the characteristic absorbance of the uranyl ion at 400–475 nm, no evidence of reduction was observed in the UV–vis spectra.⁸⁶ However, the reaction solution rapidly precipitated a yellow microcrystalline phase. An early time point reveals the uranyl band is shifted 9–14 nm from the position of free uranyl in the buffer, indicating complexation and the absorbance intensity is almost tripled (Figure S21). Single crystals were isolated by aging more dilute reaction conditions, (i.e., 0.1 mM U with 2 equivalents of PW_{11}

for 11 days). The square plate crystals (Figure S22C) match the reported structure of a autunite-type uranyl phosphate sheet with the formula $(\text{H}_3\text{O})(\text{UO}_2\text{PO}_4)(\text{H}_2\text{O})_3$ (Figure S22B).⁸⁷ Interestingly, the previous synthesis by Wang et al.⁸⁷ required 3 days of hydrothermal heating at 180 °C, in contrast to our room-temperature crystallization. However, many autunite-type materials are traditionally synthesized by room-temperature vapor diffusion,⁸⁸ so the hydrothermal approach is the exception. Raman analysis on the powder and single crystals supports that both phases are the same (Figure S22A). The main features are the uranyl and phosphate stretches at 830 and 989 nm, respectively. It was interesting to observe how quickly the uranyl ions extracted the phosphates from the POM, yet unsurprising given the historical precedent for actinide-phosphate complexation in separations chemistry (e.g., tributyl phosphate in the PUREX process) and the known low solubility of actinide-phosphate phases.^{18,89} From the conversion of the PW_9 precursor to PW_{11} in solution, there are approximately 0.36 U equivalents of free phosphate in the system available for early complexation (eq 5). However, the extent of the uranyl depletion indicates further phosphate extraction from PW_{11} . The PW_{11} -mediated reduction in the presence of neptunyl(V) but not uranyl(VI) likely proceeds because of a one-electron process (compared to two-electron process for UO_2^{2+}) and the relative instability of the neptunyl bond compared to the uranyl bond.⁹⁰

In summary, while PW_{11} spontaneously reduces neptunyl(V) to produce a soluble Np^{4+} - PW_{11} complex, the same reaction with uranyl(VI) yields a system where uranyl reacts with trace phosphate ions (coming from the *in situ* formation of PW_{11} from the PW_9 precursor) and precipitates as uranyl phosphate while PW_{11} serves a phosphate reservoir to continue the precipitation reaction. In other words, the behavior of neptunyl(V) and uranyl(VI) (the two most common ions for Np and U) are very distinct regarding their reaction with PW_{11} . Such divergent behaviors could be the basis for future strategies to separate U and Np via selective precipitation. We envision a stepwise precipitation scheme (Figure S23) in which the PW_{11} introduction induces uranyl precipitation followed by Cs^+ addition to precipitate Np as the $\text{Np}(\text{PW}_{11})_2$ complex. We also envision that UO_2^{2+} could be kept in solution while Np^{4+} is precipitated if other Keggin POMs, that do not contain phosphate, are used (e.g., $\text{BW}_{11}\text{O}_{39}^{9-}$, $\text{GaW}_{11}\text{O}_{39}^{9-}$, $\text{SiW}_{11}\text{O}_{39}^{8-}$, or $\text{GeW}_{11}\text{O}_{39}^{8-}$). These applied extensions of the present study are underway but these existing results show promise for mixed actinide streams. However, in focusing on neptunium, the predominance of the highly mobile NpO_2^+ species in environmentally relevant conditions is especially concerning considering known contamination sites (e.g., the Savannah River Site).^{91,92} Thus, mitigating neptunium transport warrants investigation into immobilization options and POMs may become a potential solution as fully inorganic and radiation-resistant precipitants. The PW_{11} -mediated NpO_2^+ reduction to Np^{4+} and subsequent precipitation with counterions is a simple and accessible approach to neptunium capture from aqueous solution.

CONCLUSIONS

This study represents a rare example of a transuranic-POM complex that enables structural and multispectroscopy characterization (SC-XRD, solution-state and solid-state UV–vis absorbance, ^{31}P NMR, relaxometry, and Raman) on sub-100 μg quantities of actinide. The Np-PW_{11} system

facilitates relatively rapid *in situ* reduction from neptunyl(V) stock solution, bypassing the need to otherwise reduce neptunyl prior and combat hydrolytic mechanisms that usually dominate Np^{4+} chemistry in solution. A suite of techniques provides evidence for the formation of a strong $[\text{Np}(\text{PW}_{11}\text{O}_{39})_2]^{10-}$ complex, that is water-soluble, air-stable, and remains intact for at least several weeks and over a broad pH range. The $\text{Np}-\text{PW}_{11}$ system allows for the costabilization of the α and β isomers of the Keggin ion in $\text{Np}(\text{PW}_{11})_2$, a first for actinide POMs. Solid-state data captures a snapshot of the dynamic exchange between rotational isomers in solution. The behavior of neptunium in the presence of PW_{11} is unique and is drastically different from the behavior of other tetravalent metals and even uranyl. The spontaneous and easy reduction of NpO_2^+ to Np^{4+} by PW_{11} also paves the way for potential new strategies to reduce, precipitate, and eventually immobilize neptunium in certain nuclear wastes. Future studies are ongoing to extend this work to other POMs and other actinides.

■ ASSOCIATED CONTENT

Data Availability Statement

The data supporting this article have been included as part of the [Supporting Information](#). The crystal structures of $\text{Np}-\text{PW}_{11}$, $\text{Zr}-\text{PW}_{11}$, and $\text{Hf}-\text{PW}_{11}$ reported here have been deposited in the Cambridge Crystallographic Data Center under CSD numbers 2404538, 2326541, and 2421131, respectively. A preliminary version of this study was released as a preprint.⁹³

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.5c02949>.

Compilation of neptunium POM compounds previously characterized via SC-XRD; compilation of solution-state studies previously performed on neptunium POMs; additional characterization results (UV-vis, ^{31}P NMR, Raman, etc., for $\text{Np}(\text{PW}_{11})_2$); uranium experiments; crystallographic details of $\text{Np}(\text{PW}_{11})_2$, $\text{Zr}(\text{PW}_{11})_2$, and $\text{Hf}(\text{PW}_{11})_2$ (PDF)

Accession Codes

Deposition Numbers 2326541, 2404538, and 2421131 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe [Access Structures service](#).

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Author Contributions

A.M.H.: Investigation, validation, formal analysis, writing—original draft, and visualization; I.C.: investigation and formal analysis; D.C.K.: investigation and formal analysis; C.A.C.: investigation; G.J.-P.D.: conceptualization, methodology, resources, writing—original draft, writing—review and editing, supervision, and funding acquisition.

Notes

The authors declare no competing financial interest.

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